Lecture 2

Functions and the real numbers

Uses.

Summary 1: Sets specify domains. An equivalence requires both implications. Quantifiers specify the order of choices. The number $\sqrt{2}$ is irrational.

Objectives. *Limits concern the values of functions and the order of real numbers. We specify the functions we use, prove the estimates needed for sequence limits and state completeness. Completeness will supply an integer beyond any prescribed real bound.*

By the end of this lecture you will be able to:

- Specify a function's domain, codomain and image and check an inverse.
- Prove Bernoulli's inequality by induction and use absolute-value bounds.
- Distinguish a least upper bound from a largest element.
- Derive the Archimedean property from completeness.

2.1 Functions and their images

A *function* $f : A \to B$ assigns to each $x \in A$ exactly one element $f(x) \in B$. The set A is its *domain*. The set B is its *codomain*. The values actually attained form its *image*, written

$$f(A) = \{f(x) : x \in A\} \subseteq B.$$

Specifying a formula without a domain does not specify the whole function. The codomain matters when we ask whether every possible output is attained.

For example, let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Its image is $[0, \infty)$. Every square is nonnegative. Conversely, if $y \geq 0$, the real number $x = \sqrt{y}$ satisfies $f(x) = y$. These two implications prove the image assertion. The function does not attain a negative value even though its codomain contains such values.

The *graph* of $f : A \to B$ is the set of input-output pairs

$$\{(x, f(x)) : x \in A\} \subseteq A \times B.$$

Here $A \times B$ denotes the set of ordered pairs with first entry in A and second entry in B. A graph records which output belongs to each input. It does not replace the specification of the codomain.

A function $f : A \to B$ is *injective* if equal outputs imply equal inputs. It is *surjective* if its image equals B. It is *bijective* if it has both properties. In symbols, injectivity is

$$\forall x_1, x_2 \in A : \quad f(x_1) = f(x_2) \implies x_1 = x_2.$$

Surjectivity is

$$\forall y \in B \quad \exists x \in A : \quad f(x) = y.$$

The order of quantifiers in the second statement permits the input x to depend on the requested output y.

The squaring function $f : \mathbb{R} \to \mathbb{R}$ is not injective, because $f(1) = f(-1)$. It is not surjective, because -1 is not attained. Changing its codomain to $[0, \infty)$ makes it surjective. Changing its domain to $[0, \infty)$ as well makes it bijective. To check injectivity on this domain, suppose $x_1^2 = x_2^2$. Then $(x_1 - x_2)(x_1 + x_2) = 0$. If $x_1 + x_2 = 0$, nonnegativity gives $x_1 = x_2 = 0$. Otherwise $x_1 - x_2 = 0$. In either case the inputs are equal.

2.2 Composition and inverses

Let $f : A \to B$ and $g : B \to C$. Their *composition* $g \circ f : A \to C$ is given by

$$(g \circ f)(x) = g(f(x)).$$

The function f is applied first. Its output is an allowed input for g. More generally, composition is defined whenever the image of f is contained in the domain of g. This domain condition is part of the operation.

The *absolute value* of a real number is its distance from zero. It is given by $|x| = x$ for $x \geq 0$ and $|x| = -x$ for $x < 0$. Thus $|x| \geq 0$, and $|x| = 0$ exactly when $x = 0$. We have $|xy| = |x||y|$ and $|-x| = |x|$, by these sign cases.

For example, let $f : \mathbb{R} \to [0, \infty)$ be given by $f(x) = x^2$ and let $g : [0, \infty) \to [0, \infty)$ be given by $g(y) = \sqrt{y}$. Their composition on $\mathbb{R}$ is

$$(g \circ f)(x) = \sqrt{x^2} = |x|.$$

On $[0, \infty)$, the composition in the opposite order satisfies $(f \circ g)(y) = y$. The order and domain both matter. A calculation of $g(f(x))$ does not determine $f(g(y))$ on an unspecified domain.

An *inverse function* of $f : A \to B$ is a function $f^{-1} : B \to A$ satisfying

$$f^{-1}(f(x)) = x \quad (x \in A), \qquad f(f^{-1}(y)) = y \quad (y \in B).$$

A function has an inverse exactly when it is bijective. Indeed, a bijection supplies for each $y \in B$ one and only one input $x \in A$. Assigning that input to y defines its inverse. Conversely, applying an inverse to an equality of outputs proves injectivity. The identity $f(f^{-1}(y)) = y$ proves that each $y \in B$ is attained, which is surjectivity.

For the squaring function on $[0, \infty)$, the inverse is the positive square-root function. On all of $\mathbb{R}$, squaring has no inverse. The identity $\sqrt{x^2} = |x|$ shows exactly why taking a positive square root cannot recover a negative input.

Standing fact E. We use the school exponential, logarithmic and trigonometric functions with their school identities, domains, images and monotonicity on the usual restricted intervals. Real powers with positive bases have their school power laws. Exponentials a^x for $a > 0$ have their school order properties. The logarithm ln is the inverse of the exponential with base e, whose real number e will be constructed in Lecture 4. The trigonometric functions use radians. One of the assumed estimates is $\cos t \leq (\sin t)/t \leq 1$ for $0 < |t| < \pi/2$. The continuity assumptions for these functions will be stated with the definition of continuity

in Lecture 5. These facts are assumptions of the course. A later power-series argument will identify the exponential, sine and cosine with their series.

For reference, the exponential e^x has domain $\mathbb{R}$ and image $(0,\infty)$. Its inverse logarithm has domain $(0,\infty)$ and image $\mathbb{R}$. Sine and cosine have domain $\mathbb{R}$ and image $[-1,1]$. Tangent is defined where cosine is nonzero.

The *hyperbolic sine* and *hyperbolic cosine* are functions $\mathbb{R}\to\mathbb{R}$ defined by

$$\sinh x=\frac{e^x-e^{-x}}{2},\qquad \cosh x=\frac{e^x+e^{-x}}{2}.$$

They are combinations of exponentials.

Standing fact E includes strict monotonicity and the images of the following restrictions. Sine on $[-\pi/2,\pi/2]$ is strictly increasing onto $[-1,1]$. Cosine on $[0,\pi]$ is strictly decreasing onto $[-1,1]$. Tangent on $(-\pi/2,\pi/2)$ is strictly increasing onto $\mathbb{R}$.

Inverse trigonometric functions likewise require restricted domains. We define $\arcsin:[-1,1]\to[-\pi/2,\pi/2]$ as the inverse of sine restricted to $[-\pi/2,\pi/2]$. We define $\arccos:[-1,1]\to[0,\pi]$ using cosine on $[0,\pi]$, and $\arctan:\mathbb{R}\to(-\pi/2,\pi/2)$ using tangent on that open interval. For example, $\arcsin(\sin\pi)=0$, which is different from π. The inverse identity is asserted only on the restricted domain.

2.3 Induction and Bernoulli's inequality

Mathematical induction proves a property $P(n)$ for every $n\in\mathbb{N}$ by establishing a starting case and an implication from one index to the next. The principle asserts that if $P(1)$ holds and $P(n)\Rightarrow P(n+1)$ for every $n\in\mathbb{N}$, then $P(n)$ holds for every $n\in\mathbb{N}$. We use induction as a basic property of the natural numbers.

The induction hypothesis is a statement at an arbitrary fixed index n. It is an assumption used to prove the implication to $n+1$. It is not an assumption that the final conclusion already holds at every index. The starting case is needed to apply the successive implications.

Theorem 2.1 (Bernoulli's inequality). *For every $h\in\mathbb{R}$ with $h\ge -1$ and every $n\in\mathbb{N}$,*

$$(1+h)^n\ge 1+nh.$$

Proof. Fix $h\ge -1$. We argue by induction on n. At $n=1$ the assertion is equality. Suppose it holds at n. Since $1+h\ge 0$, multiplication by $1+h$ preserves the inequality. Expansion gives

$$(1+h)^{n+1}\ge (1+nh)(1+h)=1+(n+1)h+nh^2\ge 1+(n+1)h.$$

The last inequality follows from $n\ge 1$ and $h^2\ge 0$. This proves the induction step and hence the theorem. ∎

The restriction $h\ge -1$ makes the multiplication in the proof valid. The factor $1+nh$ need not be positive. The proof does not multiply an inequality by that factor, so its sign does not affect the argument. The endpoint $h=-1$ is included, since its multiplier is zero.

For $h=1/2$ and $n=4$, the inequality gives $(3/2)^4\ge 3$. The exact left side is $81/16$. The bound is deliberately simpler than the expression it controls. When $h>0$, taking reciprocals of the positive quantities gives

$$\frac{1}{(1+h)^n}\le\frac{1}{1+nh}.$$

The right side will let us turn geometric decay into an explicit cutoff in the next lecture.

2.4 Absolute value and error estimates

For $r \geq 0$, the relation $|x| \leq r$ is equivalent to $-r \leq x \leq r$. This converts absolute-value bounds into ordered inequalities. Applying it to both x and y and adding the bounds gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Applying the same equivalence to the sum proves the triangle inequality.

Theorem 2.2 (Triangle inequalities). *For $x, y \in \mathbb{R}$,*

$$|x + y| \leq |x| + |y|, \qquad \big||x| - |y|\big| \leq |x - y|.$$

Proof. The first inequality was proved by the ordered bounds above. For the second, apply the first to $x = (x - y) + y$ to obtain $|x| - |y| \leq |x - y|$. Interchanging x and y gives $|y| - |x| \leq |x - y|$. Together these inequalities say that the absolute value of $|x| - |y|$ is at most $|x - y|$. ∎

For a proposed value $a \in \mathbb{R}$, the number $|x - a|$ measures the error in using x for a. The triangle inequality then bounds a total error by the sum of two intermediate errors:

$$|x - a| \leq |x - b| + |b - a| \qquad (x, a, b \in \mathbb{R}).$$

This estimate does not depend on the order of the three points on the real line. It is useful precisely when that order is not known.

2.5 Bounds and completeness

Let $A \subseteq \mathbb{R}$ be nonempty. A real number u is an *upper bound* of A if $x \leq u$ for every $x \in A$. The set is *bounded above* if an upper bound exists. A *lower bound* reverses this inequality. The set is *bounded* if it has both an upper and a lower bound. A *maximum* of A is an upper bound that belongs to A.

Equivalently, A is bounded if some $M > 0$ satisfies $|x| \leq M$ for all $x \in A$. To verify this equivalence, suppose ℓ and u are lower and upper bounds. The number $M = \max\{|\ell|, |u|, 1\}$ is positive and satisfies $-M \leq \ell \leq x \leq u \leq M$ for each $x \in A$. Hence $|x| \leq M$. Conversely, such an M supplies the lower bound $-M$ and the upper bound M.

A *supremum*, written $\sup A$, is a least upper bound. Its defining conditions, for $s \in \mathbb{R}$, are

$$x \leq s \quad \text{for every } x \in A, \qquad s \leq u \quad \text{for every upper bound } u \text{ of } A.$$

Two least upper bounds are each at most the other, so they are equal. Thus the notation $\sup A$ denotes a unique number when a least upper bound exists.

The supremum need not belong to A. Dually, the *infimum* $\inf A$ is a greatest lower bound.

For example, the interval $A = (0, 1)$ has supremum 1 and no maximum. Every member is less than 1, so 1 is an upper bound. To prove it is least, let $u < 1$. If $u \leq 0$, the member $1/2$ exceeds u. If $0 < u < 1$, the member $(u + 1)/2$ exceeds u and is less than 1. Thus no $u < 1$ is an upper bound. Finally, 1 does not belong to the interval, so it is not a maximum.

More generally, if $s = \sup A$, then for each $\varepsilon > 0$ there exists $x \in A$ with $s - \varepsilon < x \leq s$. Otherwise $s - \varepsilon$ would be an upper bound smaller than s. The element x may depend on ε. This consequence of leastness will construct limits of monotone sequences.

Standing fact R. The real numbers have the following *completeness* property: every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. We assume this property and do not construct the real numbers here. The ordinary field and order rules of school arithmetic remain in force.

Completeness does not say that every set has a largest element. The open interval just considered has a supremum without a largest element. It also does not say that an unbounded set has a finite supremum. Both nonemptiness and an upper bound are hypotheses of standing fact R.

2.6 The Archimedean property

Theorem 2.3 (Archimedean property). *For every $x \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that $N > x$.*

Proof. We argue by contradiction. Suppose $\mathbb{N}$ were bounded above in $\mathbb{R}$. By standing fact R, it would have a supremum $s \in \mathbb{R}$. Since $s - 1 < s$, the number $s - 1$ cannot be an upper bound. Choose $n \in \mathbb{N}$ with $n > s - 1$. Then $n + 1 \in \mathbb{N}$ and $n + 1 > s$. This contradicts the fact that s is an upper bound of $\mathbb{N}$. Hence $\mathbb{N}$ is not bounded above, which is the claimed property. ∎

For $\varepsilon > 0$, apply the theorem to $x = 1/\varepsilon$. It supplies $N \in \mathbb{N}$ with $N > 1/\varepsilon$, so $1/N < \varepsilon$. It supplies such an integer without requiring $1/\varepsilon$ to be an integer. The strict inequality is useful when the final error bound is also strict.

The same theorem implies that every real x lies between two consecutive integers. Choose $K \in \mathbb{N}$ with $K > |x| + 1$. Among the finitely many integers from $-K$ to K, choose the largest m with $m \le x$. This set of choices is nonempty because $-K < x$. Since $m \le x < K$ and m and K are integers, we have $m + 1 \le K$. If $m + 1 \le x$, then $m + 1$ would be another admissible integer in the finite search, contradicting maximality. Hence $x < m + 1$. Thus $m \le x < m + 1$. We call m the *floor* of x and write $m = \lfloor x \rfloor$.

It follows that between any two real numbers $a < b$ there is a rational number. Choose $N \in \mathbb{N}$ with $N > 1/(b - a)$ and set $m = \lfloor Na \rfloor + 1$. Then $Na < m \le Na + 1 < Nb$. Division by $N > 0$ gives

$$a < \frac{m}{N} < b.$$

The middle number is rational. This property is called *density of $\mathbb{Q}$ in $\mathbb{R}$*.

Density locates the failure of completeness in the rational numbers. Consider the set $S = \{q \in \mathbb{Q} : 0 < q < \sqrt{2}\}$. It is nonempty, since $1 \in S$, and is bounded above by 2. In $\mathbb{R}$, its supremum is $\sqrt{2}$: that number is an upper bound, and density supplies a member of S above any smaller proposed upper bound. For a proposed bound below or equal to zero, the member 1 already suffices. Since $\sqrt{2}$ is irrational by Theorem 1.4, it cannot be a rational least upper bound. A rational upper bound strictly above $\sqrt{2}$ is not least, because density supplies a smaller rational upper bound between it and $\sqrt{2}$. Thus S has no supremum in $\mathbb{Q}$.

Summary 2.

- A function includes its domain and codomain. Its image consists of attained values. An inverse exists exactly for a bijection.
- Theorem 2.1: $(1+h)^n \geq 1+nh$ for $h \geq -1$ and $n \in \mathbb{N}$. The proof is by induction.
- Theorem 2.2: the triangle and reverse triangle inequalities bound sums and changes of absolute values.
- Standing fact R gives a supremum to every nonempty real set bounded above. A supremum is a least upper bound, not necessarily a maximum.
- Theorem 2.3 supplies integer cutoffs. Consequently, $\mathbb{Q}$ is dense in $\mathbb{R}$ but does not have the completeness property.

Lecture 3 uses the estimates and integer choices to prove sequence limits. The least-upper-bound property itself will establish convergence of monotone bounded sequences in Lecture 4. Sheet 2 practises functions, induction, bounds and hyperbolic branches.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §§1.1–1.2 and 2.1–2.4. S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, Chapter 1.